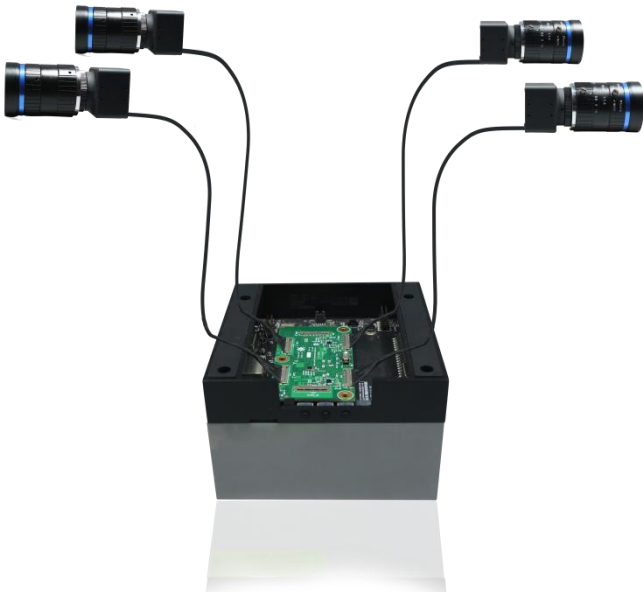


e-CAM82_CUOAGX

Release Notes



Version 1.4

e-con Systems

3/14/2023



e-con Systems

Think Camera. Think e-con.

Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

Contents

INTRODUCTION	3
RELEASE DETAILS	4
DOWNLOAD RELEASE PACKAGE	8
TROUBLESHOOTING	9
FAQ	10
WHAT'S NEXT?	11
SUPPORT	12

Introduction

e-con Systems is specialized in designing the multi-camera synchronous solution for Jetson platforms. In continuation to multi-camera solutions, e-con Systems has developed a new multi-camera board called e-CAM82_CUOAGX board. The multi-camera board can be directly interface with NVIDIA™ Jetson Orin/AGX Xavier development kit.

e-CAM82_CUOAGX board interconnects up-to four 8MP custom lens camera modules in 4-Lane and six camera modules in 2-Lane mode. It has 1/1.2" IMX485 CMOS image sensor from SONY. The 8MP color cameras are provided with CS-mount lens holder.

Release Details

- Release Version: **R02_RC1**
- Release Date: **13-MAR-2023**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.1
L4T version	L4T_R35.2.1
Kernel version	5.10.104-g018a5562a
MCU_FW version	1_2_e-CA_6_1_2_0_df25613
eCAM_argus_camera version	0.98.3:11f2a3e

- Release Version: **R01_RC5**
- Release Date: **22-FEB-2023**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
MCU_FW version	1_2_e-CA_6_1_2_0_df25613
eCAM_argus_camera version	0.98.3:546c93d

- Release Version: **R01_RC4**
- Release Date: **24-NOV-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
MCU_FW version	1_2_IMX4_6_1_1_1-42f8c51
eCAM_argus_camera version	0.98.3:546c93d

- Release Version: **R01_RC3**
- Release Date: **16-NOV-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
MCU_FW version	1_2_IMX4_6_1_1_1-42f8c51
eCAM_argus_camera version	0.98.3:546c93d

- Release Version: **R01_RC2**
- Release Date: **09-NOV-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-ge3f18e5b3
MCU_FW version	1_2_IMX4_6_1_1_1-42f8c51
eCAM_argus_camera version	0.98.3:546c93d

- Release Version: **R01_RC1**
- Release Date: **28-OCT-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-ge3f18e5b3
MCU_FW version	1_2_IMX4_6_1_1_1-42f8c51
eCAM_argus_camera version	0.98.3

- Release Version: **R01_RC1**
- Release Date: **19-JUL-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.1 DP
L4T version	L4T_R34.1.1
Kernel version	5.10.65-tegra
MCU_FW version	1_2_IMX4_6_1_1_0_95df2bc
eCAM_argus_camera version	0.98.3

- Release Version: **R01_RC1**
- Release Date: **12-JUL-2022**
- Product: **e-CAM82_CUOAGX**

Description	JETSON AGX ORIN/AGX XAVIER
Host machine version required for flashing software onto Jetson AGX Orin.	Ubuntu 18.04 or 20.04 (aarch64)
Sample rootfs Ubuntu operating system to run on Jetson AGX Orin.	Ubuntu 20.04 (aarch64)
JetPack version	JetPack 5.0.1 DP
L4T version	L4T_R34.1.1
Kernel version	5.10.65-tegra
MCU_FW version	1_2_IMX4_6_1_1_0_95df2bc
eCAM_argus_camera version	0.98.3

Download Release Package

In this section, you can get the FTP credentials for downloading Release Package.

For e-CAM82_CUOAGX,

FTP Hostname: <ftp.e-consystems.com>

FTP Username: e-CAM82-CUOAGX@ftp.e-consystems.com

Password: Cucagx8822\$#

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

Recording the video closes the sample application

This is a known limitation. Recording the video in any supported resolution more than once will close the application.

To recover the application, run the following command.

```
$ sudo service nvargus-daemon restart
```

When four cameras connected multi session mode doesn't work properly.

This is a known limitation. This limitation is because of Argus application provided with Jetpack5.1.

In any supported resolutions captured images are corrupted when output size is not 0x0

This is a known limitation. This limitation is because of Argus application provided with Jetpack5.1.

v4l2-compliance test fails in buffer ioctl

VIDIOC_REQBUFS ioctl fail in v4l2-compliance test. This fail is because of Nvidia's V4L2 framework implementation.

Streams stops randomly when running the cameras for longer days.

This is a known issue.

To recover the stream, run the following command and re-launch the application

```
$ sudo service nvargus-daemon restart
```

FAQ

1. Does e-con Systems have any plan to support longer length ipex cable?

e-con Systems provide a very flexible 30 cm ipex cable along with this kit. For customization, please write to camerasolutions@e-consystems.com with your requirement.

What's Next?

After understanding the Release Notes, you can refer to the following documents to understand more about e-CAM82_CUOAGX.

- *e-CAM82_CUOAGX Getting Started Manual*
- *e-CAM82_CUOAGX Release Package Manifest*
- *e-CAM82_CUOAGX Linux App User Manual*

Support

Contact Us

If you need any support on e-CAM82_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	09-Nov-2022	Initial Draft	Camera Dev Team
1.1	16-Nov-2022	Added support for Xavier	Camera Dev Team
1.2	22-Feb-2023	4-lane support	Camera Dev Team
1.3	10-Mar-2023	Updated Release Details section	Camera Dev Team
1.4	14-Mar-2023	Updated L4T35.2.1 version	Camera Dev Team